

OLabs Makes School Laboratories Accessible Anytime, Anywhere

School education in India faces many challenges including the lack of infrastructure and shortage of trained teachers. Even in places where this is not a major concern, students come out with little practical knowledge of the concepts and theories they learn. Read on to discover how CDAC is addressing this problem.

Much has been said about the inadequate 'higher order thinking skills' among students passing out from our education system. As per the curriculum, science subjects should have associated labs, the study of mathematics should be accompanied with activities, and so on. But due to various constraints, these guidelines are not implemented adequately in most Indian schools. Lab access is usually limited in terms of time and resources available. And in many places, there is no lab worth mentioning. Added to this is the need to continuously replenish stock of raw materials to conduct the lab activities.

Virtual labs address this deficiency in practical exposure. These labs simulate the physical processes on a computer system, allowing the student to perform the activity on a computer screen. For example, in a chemistry virtual lab, the student can put a beaker on the table, add some chemicals to it, mix it with a spoon, boil it, and so on—all while sitting in front of the computer. As a physics student, you can suspend a spring on a stand, add various weights to it, observe it oscillating and coming to rest, measure the increase in the spring's length, and plot a graph to verify Hooke's Law. Online Labs (OLabs) is a set of such virtual labs, for schools.

Computers are today pretty accessible and affordable for students. Getting Indian languages on them is also easy. Fast-dropping prices and healthy competition is helping matters further. Today, the trend is to use tablets instead of laptops and desktops, which can be considered as a computer for most purposes. One major strength of these devices is that they are used in a personal way, unlike, for example, a TV. And, hence, students can spend as much or as little time as they need on

their topics, without worrying about other students in the class. Therefore, a computing device offers tremendous scope for open-paced learning, to account for varying learner profiles. OLabs exploits this

opportunity, offering a number of virtual labs for schools, allowing students to practice on them as and when they are comfortable, at their own pace. The project is supported by the Department of Electronics and Information Technology (DeitY), government of India, and designed and implemented by CDAC Mumbai and Amrita Vishwa Vidyapeetham.

The OLabs approach

If you do a Google search for any specific topic, say Hooke's Law, chances are that you will come across plenty of material. Good animations and even simulations can be located. Many of these are available for free, at least for use. But these are hardly used in any schools. There are two major reasons for this. First, for such online resources to be used in a school, these first need to be studied well and integrated into the curriculum. This implies

that the online resources should use the same jargon and context that children are familiar with from their curriculum. Many of the simulations do not easily fit into this mould. Teachers will need to demonstrate the concepts in class, and perhaps give assignments or lab work using the online tool. This is usually a hard task, since the tool may not be developed for such use cases. Teachers, therefore, need to understand such tools, and figure out how best they can be used in the class. There are also issues linked to fitting

